Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Methylene (CH2)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 3
Total e-   : 8
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -38.92428881
E1 (Ha) : -38.87208522
Gap (Ha) : 0.05220359
HF (Ha) : -38.92132899

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 1
Frozen e-   : 2
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : 2.000000
S   : 1.0000
<Sz>   : 0.000000
Multiplicity : 3
Spin state   : Triplet (S=1)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.997965
<D> / site : 0.399593

THIS PDF

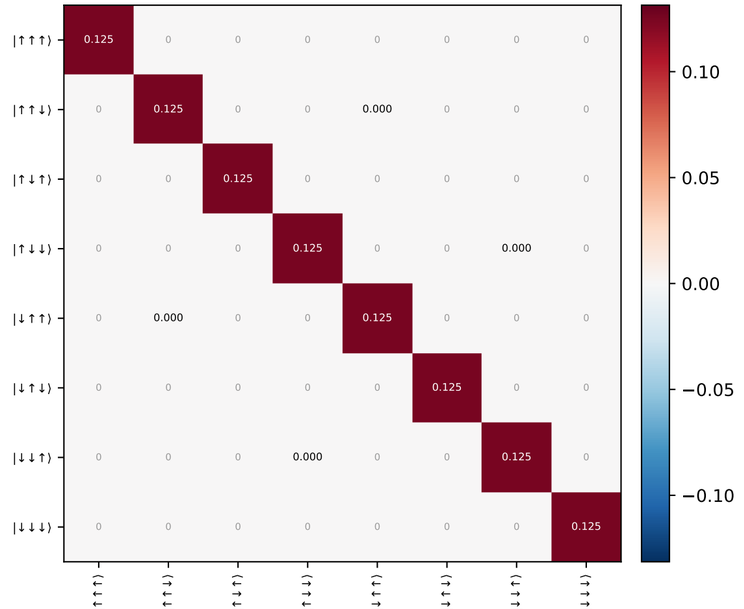
Body order : 3
Clusters   : 10

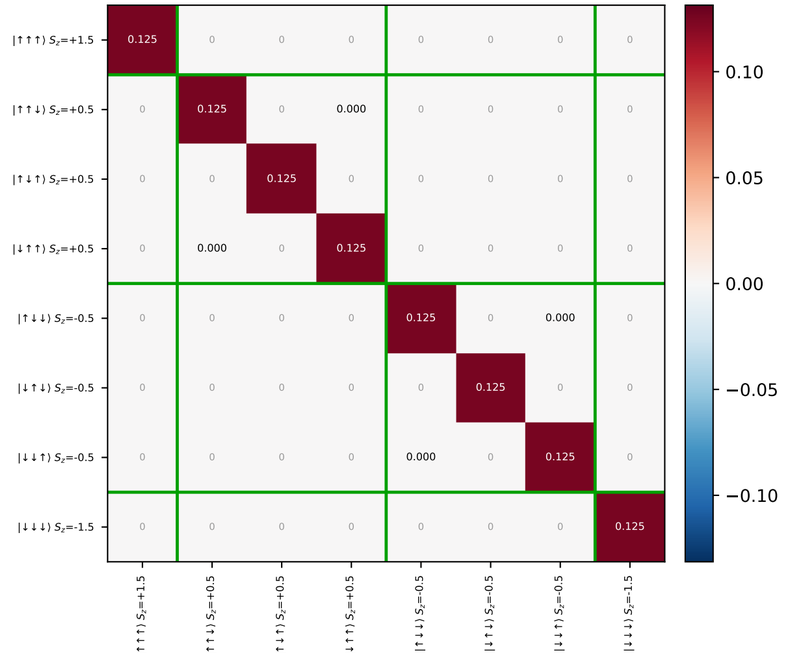
Date: 2026-07-15 09:32

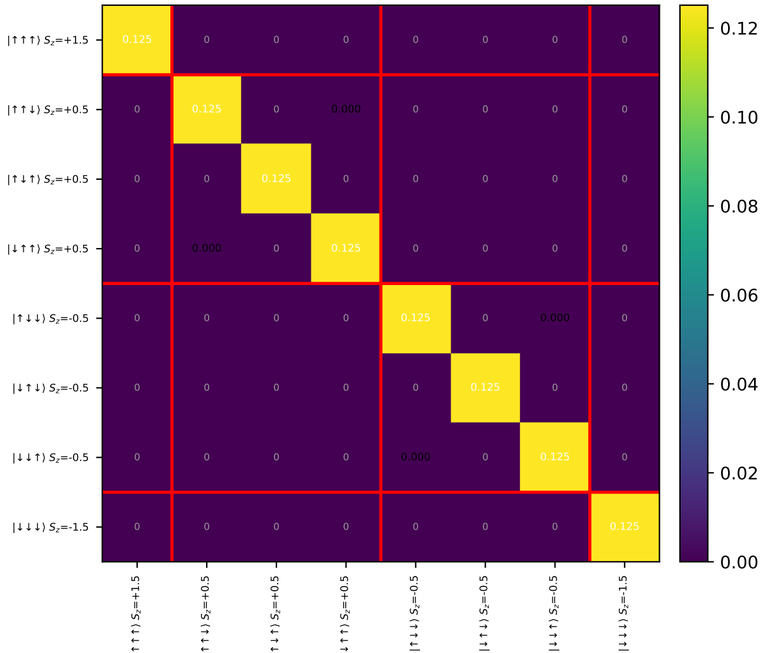
Summary — 3-Body RDMs (Methylene, CAS(6,5))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.125000	2.079441	—	—	—	—	STRONGLY MIXED
0,1,3	0.125000	2.079441	—	—	—	—	STRONGLY MIXED
0,1,4	0.179494	1.834056	—	—	—	—	STRONGLY MIXED
0,2,3	0.498930	0.701770	—	—	—	—	MIXED
0,2,4	0.268683	1.605386	—	—	—	—	MIXED
0,3,4	0.251866	1.654117	—	—	—	—	MIXED
1,2,3	0.497966	0.709751	—	—	—	—	MIXED
1,2,4	0.166176	1.915503	—	—	—	—	STRONGLY MIXED
1,3,4	0.201332	1.800221	—	—	—	—	MIXED
2,3,4	0.249009	1.448146	—	—	—	—	MIXED

3-Body RDM (Sites 0,1,2) — same matrix in all three views

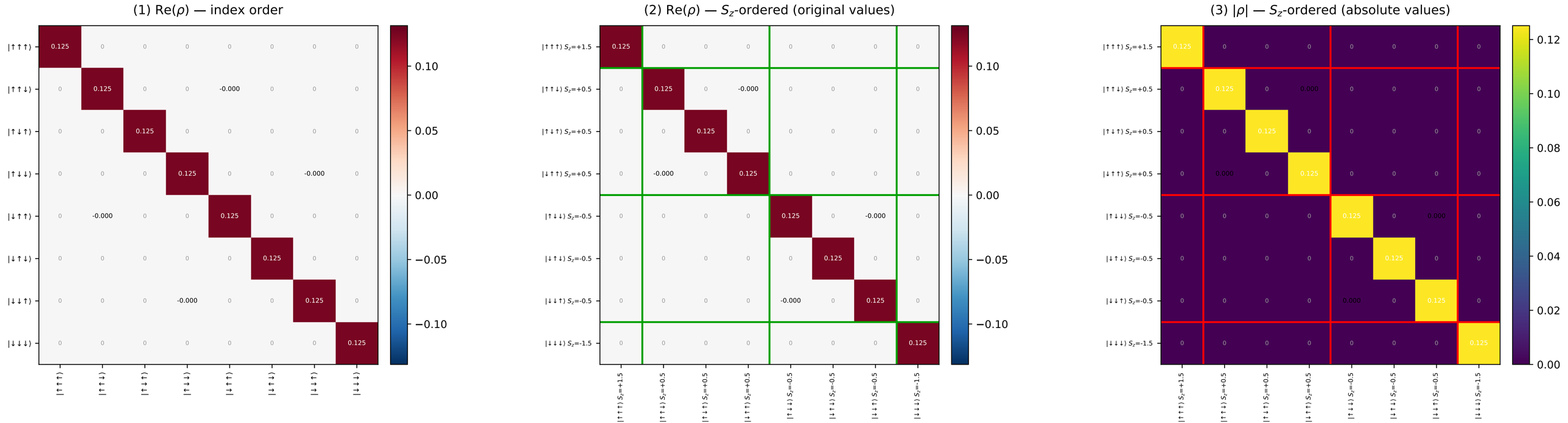






Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

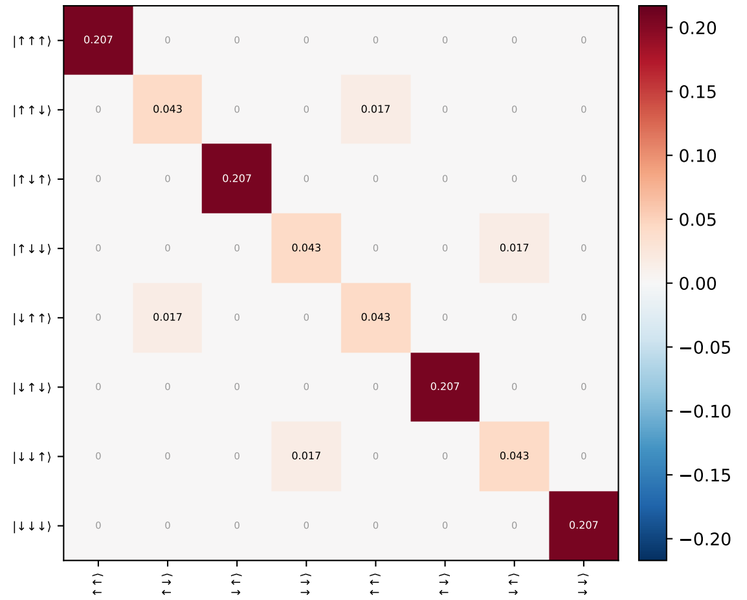
3-Body RDM (Sites 0,1,3) — same matrix in all three views



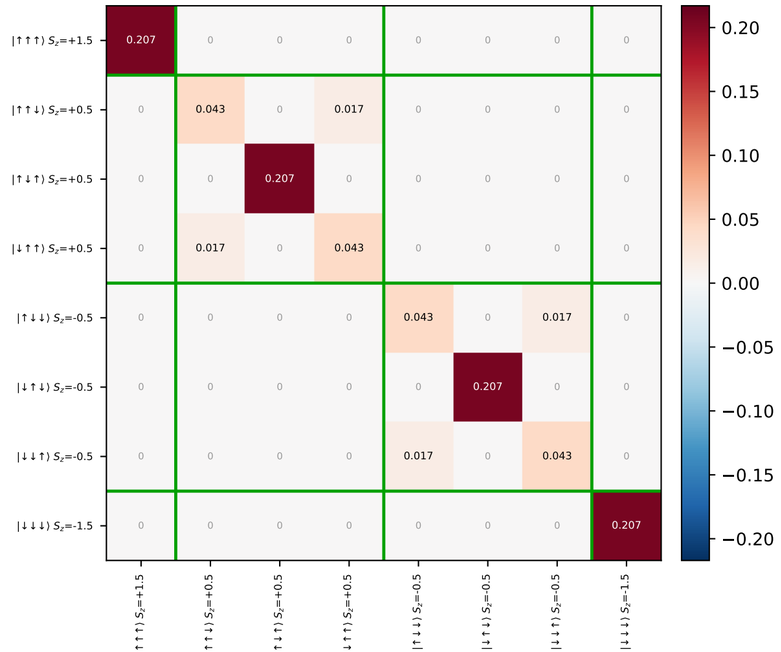
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

3-Body RDM (Sites 0,1,4) — same matrix in all three views

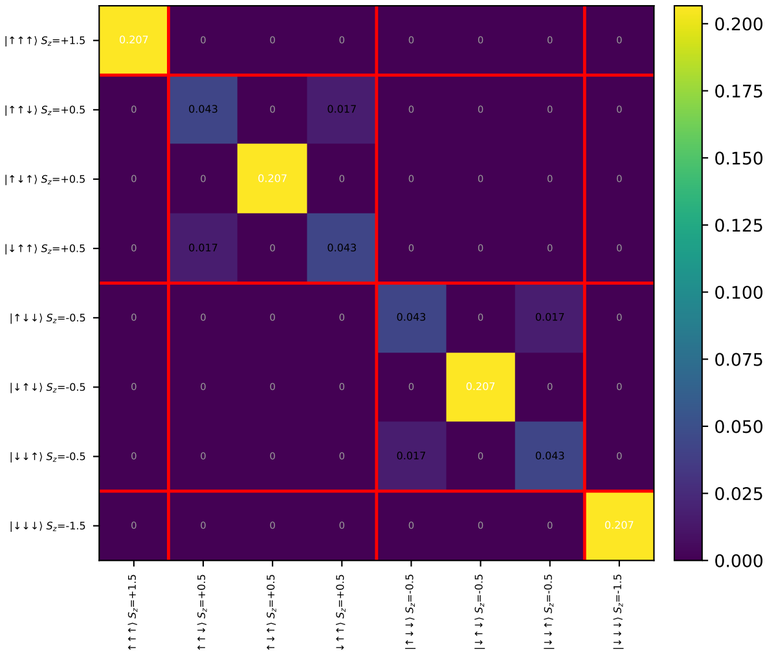
(1) Re(ρ) — index order



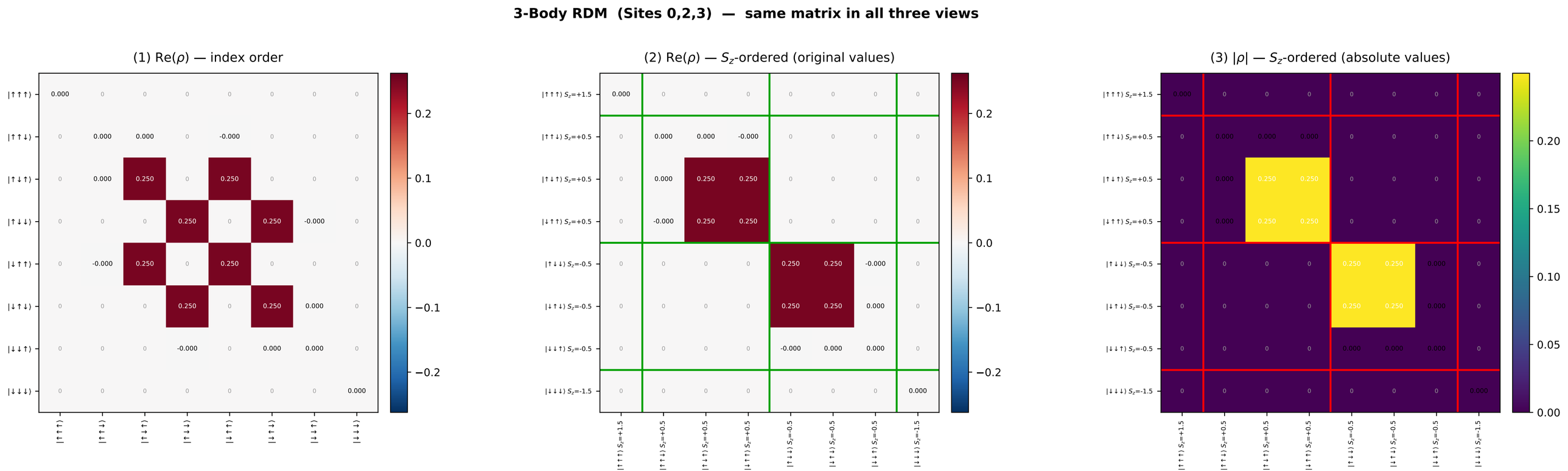
(2) Re(ρ) — S_z-ordered (original values)



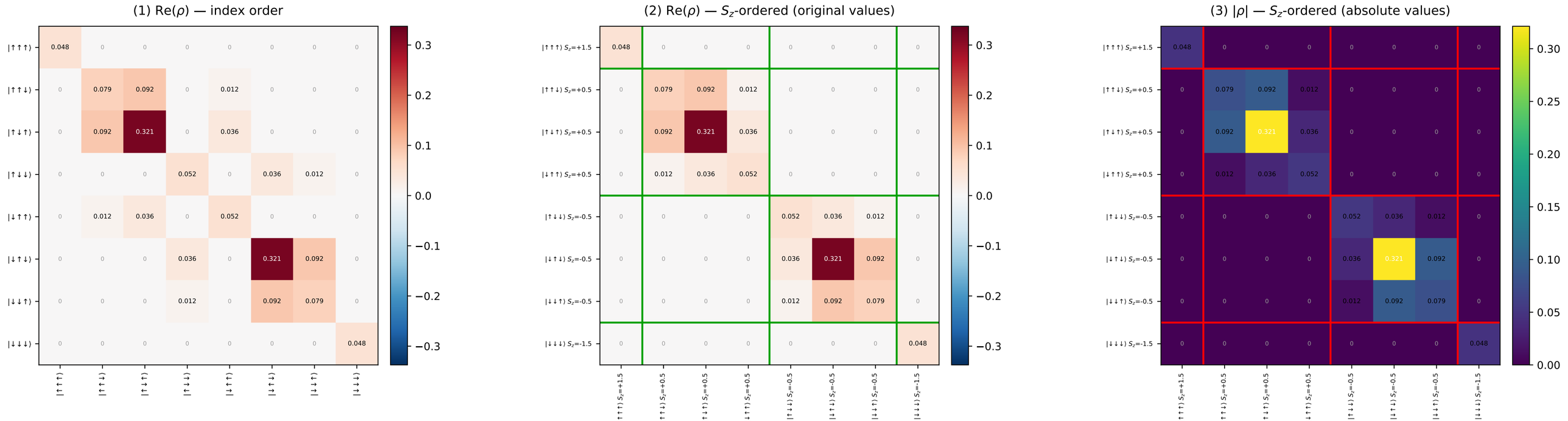
(3) $|\rho|$ — S_z-ordered (absolute values)



Trace=1.000000 Purity=0.179494 Entropy=1.834056 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

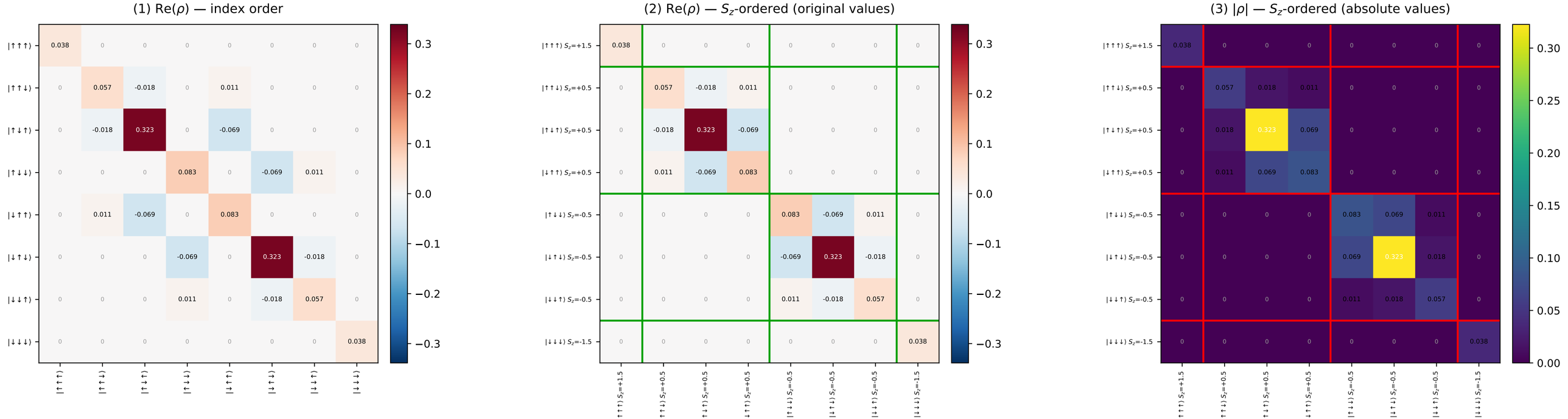

3-Body RDM (Sites 0,2,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
|↑↑↑⟩ S_z=+1.5
|↑↑↓⟩ S_z=+0.5
|↑↓↑⟩ S_z=+0.5
|↓↑↑⟩ S_z=+0.5
|↑↓↓⟩ S_z=-0.5
|↓↑↓⟩ S_z=-0.5
|↓↓↑⟩ S_z=-0.5
|↓↓↓⟩ S_z=-1.5
Trace=1.000000 Purity=0.498930 Entropy=0.701770 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

3-Body RDM (Sites 0,2,4) — same matrix in all three views



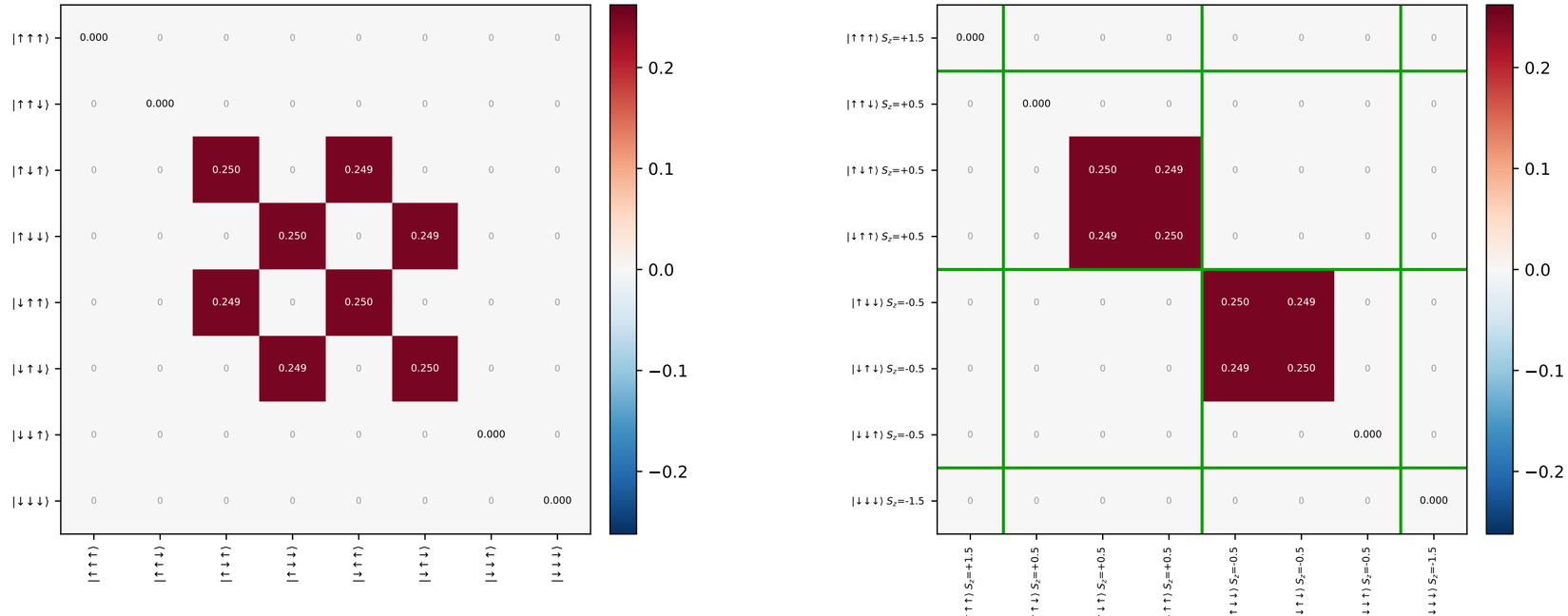
Trace=1.000000 Purity=0.268683 Entropy=1.605386 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

3-Body RDM (Sites 0,3,4) — same matrix in all three views

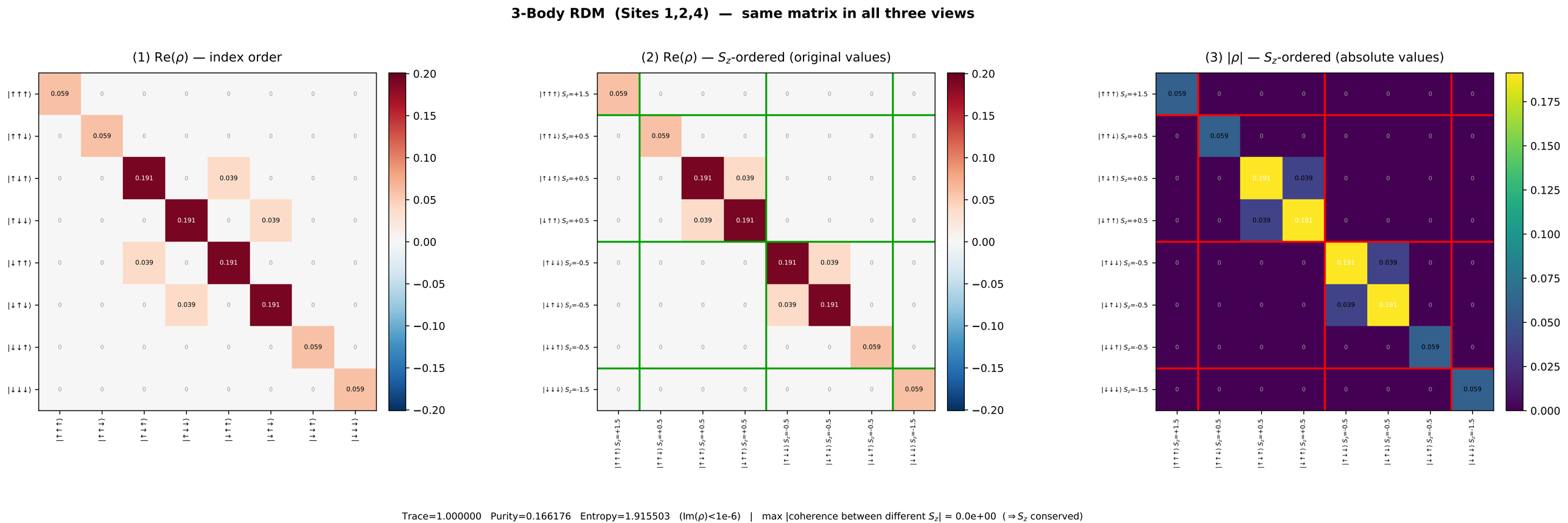


Trace=1.000000 Purity=0.251866 Entropy=1.654117 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

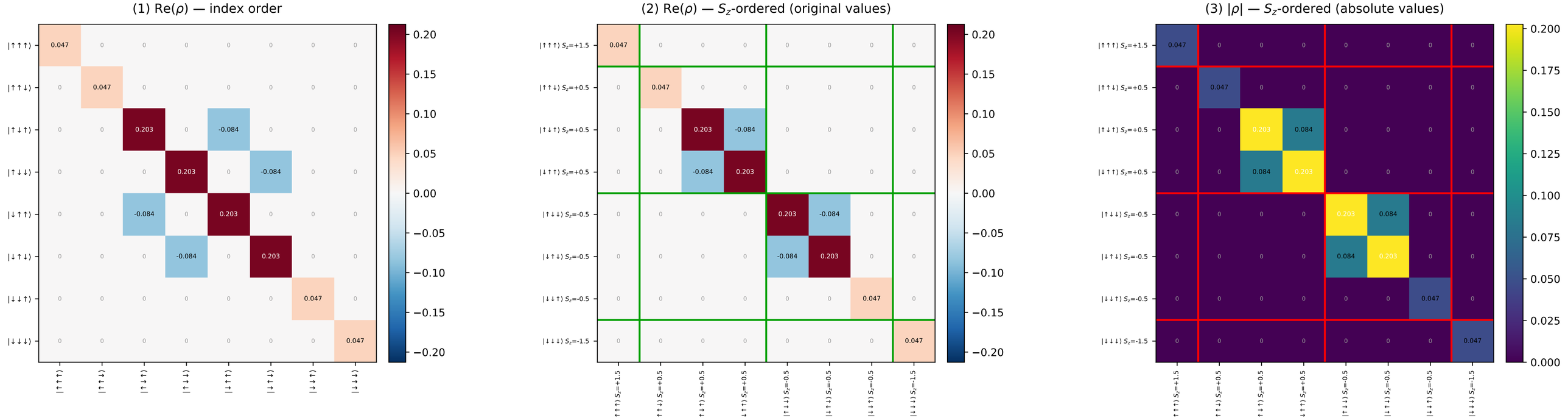
3-Body RDM (Sites 1,2,3) — same matrix in all three views



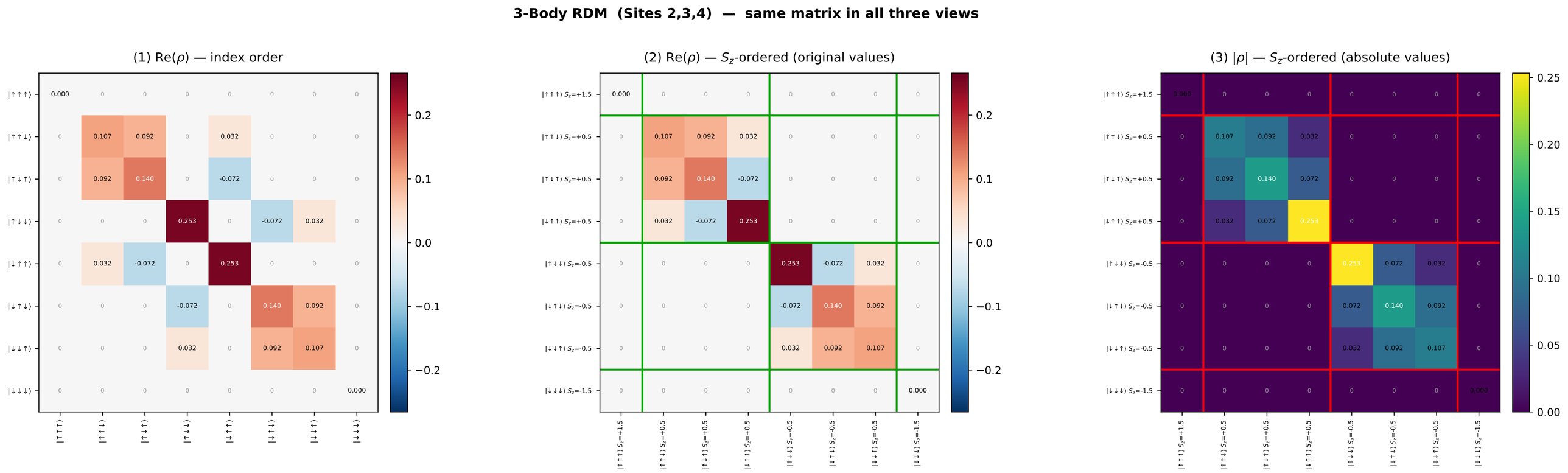
Trace=1.000000 Purity=0.497966 Entropy=0.709751 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 1,2,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.166176 Entropy=1.915503 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

3-Body RDM (Sites 1,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.201332 Entropy=1.800221 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


3-Body RDM (Sites 2,3,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑↑⟩ |↑↑↓⟩ |↑↓↑⟩ |↑↓↓⟩ |↓↑↑⟩ |↓↑↓⟩ |↓↓↑⟩ |↓↓↓⟩
|↑↑↑⟩ S_z=+1.5
|↑↑↓⟩ S_z=+0.5
|↑↓↑⟩ S_z=+0.5
|↑↓↓⟩ S_z=-0.5
|↓↑↑⟩ S_z=+0.5
|↓↑↓⟩ S_z=-0.5
|↓↓↑⟩ S_z=-0.5
|↓↓↓⟩ S_z=-1.5
Trace=1.000000 Purity=0.249009 Entropy=1.448146 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)